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18

Biologically Grounded Drug Representations for Generalizable Drug–Drug Interaction Prediction in Previously Unseen Drugs

A drug-holdout, falsification-driven investigation of biological identity and annotation popularity

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Project: DDI-Net / BIO-GINE
Submission date: 5 October 2026

1. Research Problem and Need for the Study

Drug–drug interactions (DDIs) occur when one medicine alters the effect, exposure, metabolism, or handling of another. This problem becomes increasingly important when several medicines are used together, because the number of possible drug pairs grows quadratically. Direct experimental or clinical characterization of every possible pair is therefore infeasible, which creates a role for computational screening [1], [5].

However, strong benchmark scores do not necessarily show that a model can generalize to a drug it has never encountered. Many DDI benchmarks split data at the level of interaction pairs. Under such a split, the same drug may appear in both training and test pairs, allowing a model to exploit familiar drug identities or known interaction neighborhoods instead of learning transferable properties [2]–[4].

A second methodological problem concerns biological annotations. Drug targets, enzymes, transporters, carriers, and pathways may contain transferable biological information, but annotation quantity is itself a potential shortcut. Widely studied drugs may have both more biological annotations and more documented interactions. A model may therefore appear to learn meaningful biology while mainly learning annotation popularity.

The study addresses these problems jointly: it evaluates drugs—not merely pairs—under a drug-holdout design and tests whether biological identity contributes information beyond molecular structure and beyond the number of available annotations.

2. Research Question, Objectives and Hypotheses

Central question. Can biologically grounded protein- and pathway-level representations improve drug–drug interaction prediction under drug-holdout generalization, and does any improvement reflect biological identity rather than annotation-popularity shortcuts?

- Develop BIO-GINE, a multimodal model combining a molecular graph encoder with per-drug protein and pathway encoders.
- Evaluate performance when one or both test drugs were absent from training, without introducing DDI edges into biological representations.
- Test shortcut explanations with annotation-count baselines, a degree-preserving biological-identity shuffle, and aggregation controls.
- Quantify uncertainty with paired multi-seed statistics, multiplicity correction, and calibration.

H1. True biological identity improves held-out-drug AUPRC over an aligned molecular-only model.

H2. BIO-GINE is more robust than a known-DDI-network model when both drugs are unseen (S3).

H3. Degree-preserving shuffling of biological identity reduces performance.

H4. Annotation counts alone cannot explain the performance of the full model.

3. Existing Literature and Specific Knowledge Gap

Structure-focused methods such as DeepDDI and SSI-DDI use chemical representations or substructure interactions to predict DDIs [9], [16]. Molecular structure remains available for a held-out drug, but chemistry alone may not represent mechanisms involving targets, enzymes, transporters, or pathways. Network-based methods instead exploit relationships among drugs, proteins, side effects, and other biomedical entities. Decagon models polypharmacy side effects using a multimodal graph [4];

EmerGNN explicitly considers emerging-drug prediction [3]. Such work demonstrates the value of graph context while highlighting the cold-start challenge when a drug lacks known training interactions.

DrugBank and ChEMBL supply drug–protein relationships and bioactivity evidence, UniProt standardizes protein identifiers, and Reactome organizes biological pathways [5]–[8]. These resources can also reflect unequal research attention. The unresolved question is therefore not simply whether biological features help, but whether they remain predictive under drug-disjoint evaluation and after a falsification test that preserves annotation quantity while disrupting biological identity. The claimed contribution is this evaluation-and-control design, not the invention of GINE or Deep Sets.

4. Data, Task and Leakage-Resistant Evaluation

The DDI universe is derived from the Therapeutics Data Commons DrugBank task [1], [5]. The frozen working subset contains 1,705 drugs and 191,392 documented interaction pairs. Source interaction types are collapsed to a binary documented-interaction target; severity, direction, and clinical importance are not predicted. The source contains documented positives but no confirmed negatives. Sampled unlabeled pairs are therefore comparison examples, not established non-interactions.

Drugs, rather than pairs, are partitioned: 1,195 training drugs, 255 validation drugs, and 255 test drugs. The pooled held-out evaluation comprises S2 pairs (exactly one unseen drug) and S3 pairs (both drugs unseen). The strict S3 setting removes known training-DDI neighborhoods from both endpoints. The reported pooled evaluation contains 84,690 test pairs, including 7,758 S3 pairs. These counts and the sampling scheme are reported from the author-reviewed research materials.

Sampled-unlabeled comparisons use degree matching based on training-pair degree only; S3 comparison pairs likewise have two unseen endpoints. Leakage auditing checks that train and test drug identities are disjoint, that test labels do not enter feature construction, and that biological

feature construction does not use DDI labels or split assignments. Because the evaluation uses sampled comparisons with an approximately balanced constructed prevalence, AUPRC must not be interpreted as real-world positive predictive value.

5. BIO-GINE Model and Falsification Controls

BIO-GINE combines three per-drug modalities. A GINE encoder represents the molecular atom–bond graph. A permutation-invariant Deep Sets encoder summarizes drug–protein relations while retaining relation type (target, enzyme, transporter, or carrier). A second Deep Sets encoder summarizes Reactome pathway membership. Fused drug embeddings enter a symmetric pair decoder, enforcing $f(A,B) = f(B,A)$ by construction [10]–[12]. Mean aggregation is used for the primary biological encoders to reduce direct exposure of annotation count; missing modalities receive a learned missing token. No known-DDI edges enter BIO-GINE.

Comparators include aligned molecular-only GINE (M0), an aligned Dual architecture using known-DDI network context, a biological random forest, and an annotation-count-only random forest. The central falsification control rewires biological identity while preserving annotation degree. A SUM-aggregation control tests whether explicit set-size information is predictive. These controls test alternative explanations; they do not establish a causal pharmacological mechanism.

6. Statistical Analysis and Frozen Prototype Results

17 The primary endpoint is AUPRC; AUROC, Brier score, and expected calibration error are secondary [15]. Confirmatory comparisons use paired runs across five random seeds on one fixed drug partition. Reported analyses include paired differences, 95% t-based intervals, two-sided paired tests, Holm

adjustment within the hypothesis family, and effect sizes [14]. Selected contrasts use pair bootstrap. Temperature scaling is fitted on validation predictions only [13]. Five-seed variability describes training stochasticity, not variability over independent drug splits.

Model / control	Pooled AUPRC	S3 AUPRC
BIO-GINE M4 (primary)	0.8117 ± 0.0097	0.7372 ± 0.0153
Molecular GINE (M0)	0.7784 ± 0.0059	0.7145 ± 0.0065
Aligned Dual	0.7147 ± 0.0067	0.6198 ± 0.0278
Shuffled biology	0.6923 ± 0.0054	0.6305 ± 0.0105
Annotation-count RF	0.6504 ± 0.0006	not reported
M2 curated MoA	0.8269 ± 0.0060	0.7594 ± 0.0114
M4 SUM control	0.8265 ± 0.0092	0.7485 ± 0.0127

BIO-GINE M4 exceeded aligned molecular GINE by 0.0333 pooled AUPRC (reported Holm-adjusted $p = 1.98 \times 10^{-4}$). Its S3 AUPRC exceeded aligned Dual by 0.1175 (reported Holm-adjusted $p = 6.56 \times 10^{-4}$). The degree-preserving biology shuffle reduced pooled AUPRC by 0.1195 relative to M4 (reported Holm-adjusted $p = 1.83 \times 10^{-4}$). These numerical claims are reported from the author-reviewed research materials; independent replication was not performed during editorial preparation.

Important counterevidence. The evidence ladder is non-monotonic: curated M2 outperformed preregistered M4, and the SUM aggregation control also outperformed primary mean-aggregation M4. These results challenge the assumption that adding biological sources or suppressing

annotation-count signals always improves prediction. They are reported rather than hidden.

7. Contribution, Limitations and Next Validation Steps

The project contributes a reproducible framework for asking which information sources remain useful under drug holdout, and whether improvements survive controls for annotation quantity. A successful scientific outcome includes identifying failed hypotheses and alternative explanations, not merely reporting a single favorable benchmark.

Important limitations include: (i) a selected 1,705-drug subset rather than all available drugs; (ii) sampled unlabeled pairs that may include undocumented true DDIs; (iii) curation and literature biases in biological annotations; (iv) only one fixed drug partition and five training seeds; (v) no scaffold-disjoint, temporal, or independent external validation in the reported primary results; and (vi) no patient-level variables such as dose, age, renal function, genotype, or comedication context. The system does not estimate clinical severity or establish treatment safety.

Future work should prioritize scaffold-disjoint and independent/temporal validation, improved analysis of annotation quantity and quality, and pharmacology-informed error review. Such extensions must remain separate from the already completed frozen experiments. The model is intended for research prioritization, not clinical decision support.

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